Application of thermomechanical principles to the modelling of geotechnical materials

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A thermodynamic framework is presented for the plasticity modelling of geotechnical materials. The framework is capable of modelling rigorously both friction and non-associated flow, and the strong connection between these phenomena is demonstrated. The formulation concentrates on the development of constitutive models from hypotheses about the form of an energy potential and a dissipation function. The reformulation of previous work, in which the Helmholtz free energy was used, to a new approach starting from the Gibbs free energy is found to be valuable. The relationship between the new functions and classical plasticity concepts (yield surface, plastic potential, isotropic and kinematic hardening, friction, dilation) is demonstrated. Comments are made on elastic-plastic coupling. Implications of the new approach for critical state soil models are discussed.

1. Introduction

The two disciplines of plasticity theory and thermodynamics are both mature sciences, and much work has also been carried out on the influence of thermodynamic concepts on plasticity theory. The early applications of plasticity theory were to metals, but now an equally important area of application is to soils, rocks and concrete, sometimes termed 'geomaterials'. By comparison with metals, these materials differ in that they exhibit frictional behaviour, they undergo plastic changes of volume and need to be modelled by plasticity theories with 'non-associated' flow (i.e. the plastic potential is a different function from the yield surface).

The work on the influence of thermodynamics on plasticity theory has concentrated mainly on models appropriate to metals, and so has not included the above complications. The purpose of this paper is to develop plasticity theories appropriate to geomaterials within a thermodynamic context.

One approach to the thermomechanics of geomaterials is to formulate plasticity theories, employing the conventional concepts of yield surface, plastic potential and hardening function, and then to apply retrospectively the laws of thermodynamics. This can result in restrictions on the forms of the functions, but the authors find this approach both cumbersome and unrewarding.

The alternative, pursued in this Paper, is to start with thermodynamic hypotheses, and then develop plasticity theories from them. In the course of this process as few as possible restrictions are placed on the form of the functions encountered.

This alternative proves not only to achieve greater generality, but also offers significant theoretical insights. Houlsby (1981, 1982) adopted this approach to develop plasticity models for soils, but although he demonstrated that specific models could be expressed in terms of the conventional plasticity functions, the formulation relied heavily on the use of kinematic variables, with the result that many of the functions used were unfamiliar to most geotechnical engineers.

In this paper we demonstrate more generally how plasticity theories can be derived from this approach, and recast the original formulation in a form which is more accessible to geomechanics specialists. An important outcome is a demonstration that the concept of friction is inseparably linked to the occurrence of non-associated plastic flow. This has important consequences because of the loss of the many plasticity theorems, which only apply to cases of associated flow. The overall purpose is to provide a theoretical framework for use by those working in geomechanics.

Because much of the development of the formulation depends on the use of potentials, a future possibility is the establishment of new plasticity theorems applicable to cases of non-associated flow. The reason for some optimism in this area is that the use of potentials is linked to issues of convexity and maximal principles, which are important components of the plasticity theorems. Much work remains to be done in this area.

2. Thermodynamic preliminaries

General accounts of the thermomechanics of elastic/plastic materials can be found in the books by Ziegler (1983), Lubliner (1990) and Maugin (1992) and the review articles by Germain *et al.* (1983), Ziegler & Wehrli (1987) and Reddy & Martin (1994). The approach adopted by Ziegler is the closest to that used here.

We are assuming small deformation theory, so that strains are described by the small strain tensor ε_{ij}, and the deformation rate tensor is equal to its rate of change $\dot{\varepsilon}_{ij}$. Stresses are defined so that the power of deformation per unit volume is $\sigma_{ij}\dot{\varepsilon}_{ij}$. The state of a material element is defined by ε_{ij}, an internal variable α_{ij} and either temperature θ or entropy s. The internal variable is of a kinematic type. For present purposes it is sufficient to introduce a single symmetric second order tensorial kinematic internal variable (i.e. a strain-like quantity); it can frequently be identified with the 'plastic' strain. The generalization to other forms of internal kinematic variable is possible, but not pursued here.

The first law of thermodynamics can be written in a local rate form:

$$\dot{U} = \sigma_{ij}\dot{\varepsilon}_{ij} - q_{k,k} \tag{2.1}$$

where U is the internal energy per unit volume and q_i is the heat flux vector, so that the last term in (2.1) is the rate of heat supply to the material element from its surroundings. The local heat source is assumed to be zero. The equation can be established by the standard procedure of considering the energy conservation in a volume V fixed in space. The last term arises from applying the divergence theorem to the net heat flux across the boundary of V. Similarly the local form of the second law of thermodynamics is:

$$\theta\dot{s} \geqslant \theta\dot{s}^{\mathrm{r}} \equiv -\theta\left(\frac{q_k}{\theta}\right)_{,k} = -q_{k,k} + \frac{q_k\theta_{,k}}{\theta} \tag{2.2}$$

where s is the entropy per unit volume and $\dot{s}^{\mathrm{r}}$ is the reversible part of the rate of change of entropy, which is equal to the rate of entropy supply to the material element from its surroundings. Again this expression is obtained by applying the divergence theorem to the net entropy flux across the boundary of a representative volume V. The rate of entropy production within the element is the irreversible part, $\dot{s}^{\mathrm{i}}$, which by virtue of (2.2) satisfies the inequality

$$D \equiv \theta \dot{s}^{\mathrm{i}} = \theta(\dot{s} - \dot{s}^{\mathrm{r}}) \geqslant 0 \tag{2.3}$$

where D, the dissipation function, is defined by (2.3). In this form the basic thermodynamic inequality is often referred to as Planck's inequality (Truesdell 1969).

The internal energy, being a function of state, depends on strain, the internal kinematic variable and the entropy, thus $U = U(\varepsilon_{ij}, \alpha_{ij}, s)$. For pure heating ε_{ij} and α_{ij} are constant, so that $\dot{U} = (\partial U/\partial s)\dot{s}$, and from (2.1) and (2.2) it follows that also $\dot{U} = \theta \dot{s}$, hence

$$\theta = \frac{\partial U}{\partial s} \tag{2.4}$$

cf. Ziegler (1983).

Since we are concerned with isothermal deformations in which θ is constant and uniform, it would be more convenient if s were replaced by θ in the expression for the energy function. By virtue of (2.4), in which θ is seen to be the gradient of U with respect to s, this can be achieved using a Legendre transformation to interchange the roles of θ and s.

The theory of Legendre transformations is central to the theme of this paper and the relevant parts of this theory are given in the appendix. The fundamental motivation underlying a Legendre transformation is to replace a function $Z = X(x_i)(i = 1, 2, \dots, n)$ by the 'dual' function $Z = Y(y_i)$, where $y_i = (\partial X/\partial x_i)$ are the gradients of the original function. The two functions define the same surface, Σ say, in $(n + 1)$ dimensional space. In the original description Σ is defined by the set of points (X, x_i), but in the dual description, Σ is determined by its tangent hyperplanes, defined by (Y, y_i). The dual function is given by $Y = \pm(X - x_i y_i)$, the choice of sign depends on the particular physical application. The transformation is self dual (except in certain special singular cases) in the sense that X is the dual of Y and the original independent variables $x_i = \mp(\partial Y/\partial y_i)$ are the gradients of Y. Independent variables which are not changed by the transformation are referred to as passive variables. The dual function of the internal energy is the Helmholtz free energy $F(\varepsilon_{ij}, \alpha_{ij}, \theta)$, where

$$F = U - \theta s \tag{2.5}$$

and

$$s = -\frac{\partial F}{\partial \theta} \tag{2.6}$$

is the dual relation to (2.4).

Differentiating (2.5) with respect to time, using (2.1) and (2.2) and the fact that $\dot{\theta}$ and $\theta_{,k}$ are both zero in an isothermal process, we find that

$$\dot{F} = \sigma_{ij}\dot{\varepsilon}_{ij} - \theta \dot{s}^{\mathrm{i}} \tag{2.7}$$

i.e.

$$\sigma_{ij}\dot{\varepsilon}_{ij} = \dot{F} + D. \tag{2.7$'$}$$

This central result shows that in an isothermal process the power of deformation is equal to the sum of the rate of change of free energy and the dissipation. The former represents the rate of change of recoverable, though not necessarily elastic, energy whilst D is the rate at which energy is dissipated. In a plastic material energy dissipation arises from changes in the internal variables, so that D is taken to be a function of the state variables and α_{ij}, $D = D(\varepsilon_{ij}, \alpha_{ij}, \dot{\alpha_{ij}})$.

It is possible to extend the theory to the more general case in which D is also allowed to be a function of strain rate $\dot{\varepsilon}_{ij}$. The full implications of the exclusion of this possibility remain to be explored, but the main issues involved are thought to be related to two main topics. First, rigid-plastic materials (which lack internal variables) cannot be described without allowing D to be a function of $\dot{\varepsilon}_{ij}$. They can, however, be approached within the present formulation as a limiting case of elastic-plastic materials. Second, the inclusion of $\dot{\varepsilon}_{ij}$ may be necessary in the description of certain viscous effects. Whilst the following development is sufficient to discuss rate independent elastic/plastic behaviour, Ziegler's general theory also enables viscous effects to be included and hence the construction of linear and nonlinear viscous and viscoelastic as well as viscoplastic models. To describe these viscous effects, it is necessary to regard the stress as the sum of a 'quasi-conservative' part, which is determined by the free-energy potentials and a 'dissipative' stress determined by the dissipation function, which now depends on the rate of change of total strain as well as on the rate of change of the internal variable. See also Maugin (1992, ch. 2).

An alternative expression for $\dot{F}$ for an isothermal process can be obtained by direct differentiation:

$$\dot{F} = \frac{\partial F}{\partial \varepsilon_{ij}}\dot{\varepsilon}_{ij} + \frac{\partial F}{\partial \alpha_{ij}}\dot{\alpha}_{ij} \tag{2.8}$$

which on comparison with (2.7) gives

$$\sigma_{ij} = \frac{\partial F}{\partial \varepsilon_{ij}} \tag{2.9}$$

since the rates of change of the strain and internal variable tensors are independent. We denote the corresponding stress-like variable, obtained by differentiating F (with a change of sign) with respect to the internal variable by

$$\chi_{ij} = -\frac{\partial F}{\partial \alpha_{ij}}. \tag{2.10}$$

Note that the fact that χ_{ij} is a stress-like variable follows from the fact that α_{ij} is a kinematic (strain-like) variable. It follows that the dissipation function can be written

$$D \equiv \theta \dot{s}^{\mathrm{i}} = \chi_{ij}\dot{\alpha}_{ij}. \tag{2.11}$$

It is important to note that it is the 'generalized' stress, χ_{ij}, which occurs in the inner product with $\dot{\alpha}_{ij}$ in the expressions for the dissipation, not the actual stress σ_{ij}. The difference between these two stress tensors will be denoted by $\rho_{ij} \equiv \sigma_{ij} - \chi_{ij}$.

Since we are restricting attention to isothermal processes, the explicit dependence of the energy functions on θ will now be dropped. For further developments it proves convenient also to introduce the Gibbs free-energy function $G(\sigma_{ij}, \alpha_{ij})$. By virtue of (2.9) this can be obtained from the Helmholtz free-energy function $F(\varepsilon_{ij}, \alpha_{ij})$, by a partial Legendre transformation interchanging the strain and stress variables:

$$F(\varepsilon_{ij}, \alpha_{ij}) + G(\sigma_{ij}, \alpha_{ij}) = \sigma_{ij}\varepsilon_{ij} \tag{2.12}$$

where

$$\varepsilon_{ij} = \frac{\partial G}{\partial \sigma_{ij}} \quad \text{and} \quad \chi_{ij} = \frac{\partial G}{\partial \alpha_{ij}}. \tag{2.13}$$

The latter equation follows from (2.10) and the transformation property of the passive variables in the Legendre transformation (see the appendix).

The key assumption in Ziegler's approach to constitutive equations for materials is that they are fully determined by knowledge of two functions—a thermodynamic potential, such as the Helmholtz free-energy function, or any of the related potentials derived from use of Legendre transformations, and the dissipation function. From (2.11) D is seen to be a function of $\dot{\alpha}_{ij}$ as well as possibly of α_{ij} and ε_{ij}. For a rate-independent elastic/plastic deformation, D must be a homogeneous function of degree one in $\dot{\alpha}_{ij}$ since the material does not possess a characteristic time. Hence, from Euler's theorem for homogeneous functions it follows that

$$D = \frac{\partial D}{\partial \dot{\alpha}_{ij}} \dot{\alpha}_{ij}. \tag{2.14}$$

From (2.11) and (2.14) it follows that the difference between χ_{ij} and $(\partial D/\partial \dot{\alpha}_{ij})$ must be orthogonal to $\dot{\alpha}_{ij}$. Ziegler argued that this difference is in fact zero, so that

$$\chi_{ij} = \frac{\partial D}{\partial \dot{\alpha}_{ij}} \tag{2.15}$$

and χ_{ij} is normal to the level surfaces of D. This assumption has the status of a postulate, and is usually referred to as the 'orthogonality principle'; (see Ziegler (1983, ch. 15) for a full discussion, including the form of this principle for more general types of dissipation function). This postulate has been criticized on a number of grounds, many of which are answered in the article by Ziegler (1981). Our view here is that it is a weak assumption, which effectively defines a very wide class of constitutive equations, certainly a much wider class than defined by Drucker's postulate on which many plasticity theories are based. As will be seen, the postulate admits non-associated flow rules for frictional materials, but provides some predictions of the form of this non-association. As emphasized by Ziegler (1983), if it is accepted that knowledge of the function D is sufficient to determine the corresponding generalized stresses χ_{ij}, then the relation (2.15) is the only possible form, since the normal to the level surfaces is the only vector field uniquely determined by a scalar valued function D.

3. Thermodynamic potentials for elastic/plastic materials

(*a*) *Decoupled materials*

In developments of the theory of elastic/plastic materials it is common to assume at the outset that the (small) strain tensor can be regarded as the sum of 'elastic' and 'plastic' parts. However, as shown by Lubliner (1972) (see also Lubliner 1990; Reddy & Martin 1994) it is possible to deduce this decomposition formally for a material whose instantaneous elastic moduli are independent of the internal variables—a so-called 'decoupled material'.

Differentiating the first equation (2.13) with respect to time shows that the strain rate is the sum of two terms:

$$\dot{\varepsilon} = \frac{\partial^2 G}{\partial \sigma_{ij} \partial \sigma_{kl}} \dot{\sigma}_{kl} + \frac{\partial^2 G}{\partial \sigma_{ij} \partial \alpha_{kl}} \dot{\alpha}_{kl}. \tag{3.1}$$

The coefficient of the $\dot{\sigma}_{kl}$ term is the instantaneous elastic compliance, which, for a decoupled material, is independent of the internal variable α_{ij} so that $(\partial^3 G/\partial\sigma_{ij}\partial\sigma_{kl}\partial\alpha_{mn}) = 0$. It follows that the coefficient of the $\dot{\alpha}_{kl}$ term is necessarily independent of σ_{ij}. Hence, both terms can be integrated separately with respect to time to give the elastic strain

$$\varepsilon_{ij}^{\mathrm{e}} = \int_{\sigma^0}^{\sigma} \frac{\partial^2 G}{\partial\sigma_{ij}\partial\sigma_{kl}}\,\mathrm{d}\sigma_{kl} \tag{3.2}$$

where σ_{ij}^0 is the stress in the ground state, and the plastic strain

$$\varepsilon_{ij}^{\mathrm{p}} = \int_{\sigma^0}^{\sigma} \frac{\partial^2 G}{\partial\sigma_{ij}\partial\alpha_{kl}}\,\mathrm{d}\alpha_{kl}. \tag{3.3}$$

It further follows from the decoupled assumption, upon integration of the second derivatives in (3.1), that the Gibbs free-energy function must take the form

$$G(\sigma_{ij},\alpha_{ij}) = G_1(\sigma_{ij}) + A_{ij}(\alpha_{ij})\sigma_{ij} + G_2(\alpha_{ij}). \tag{3.4}$$

The cross term is hence linear in stress. Up to this point the internal variable has not been given any specific physical significance. The tensor α_{ij} could be replaced by any symmetric second-order tensor quantity derived from α_{ij} by an invertible function, without changing the formal results. Hence, we can replace the function $A_{ij}(\alpha_{ij})$ in (3.4) by α_{ij} without any loss of generality, so that now

$$G(\sigma_{ij},\alpha_{ij}) = G_1(\sigma_{ij}) + \sigma_{ij}\alpha_{ij} + G_2(\alpha_{ij}) \tag{3.5}$$

and from (2.13) the strain is

$$\varepsilon_{ij} = \frac{\partial G_1}{\partial\sigma_{ij}}(\sigma_{ij}) + \alpha_{ij} \tag{3.6}$$

i.e.

$$\varepsilon_{ij} = \varepsilon_{ij}^{\mathrm{e}}(\sigma_{ij}) + \varepsilon_{ij}^{\mathrm{p}} \tag{3.7}$$

where the elastic strain $\varepsilon_{ij}^{\mathrm{e}}$ is a function just of stress and α_{ij} can now be identified as the plastic or permanent strain, i.e. the strain remaining when the stress in the material element returns to its initial value σ_{ij}^0. With this choice of α_{ij}, the mixed second derivative $(\partial^2 G/\partial\sigma_{ij}\partial\alpha_{kl}) = \delta_{ij}\delta_{kl}$, so that (3.1) is recovered by differentiating (3.6) with respect to time. On transforming back to the Helmholtz free energy the above model could be expressed in the decomposed form $F = F_1(\varepsilon_{ij} - \alpha_{ij}) + F_2(\alpha_{ij}) = F_1(\varepsilon_{ij}^{\mathrm{e}}) + F_2(\alpha_{ij})$.

It is important to note that the function $G_2(\alpha_{ij})$ in (3.5) does not influence the elastic response. From the second equation (2.13), the generalized stress for a decoupled material must be of the form

$$\chi_{ij} = \sigma_{ij} + \frac{\partial G_2(\alpha_{ij})}{\partial\alpha_{ij}} = \sigma_{ij} - \rho_{ij}(\alpha_{ij}). \tag{3.8}$$

The G_2 function hence defines the stress already defined to be the difference between the true and generalized stress variables, and which is a function of the plastic strain. As will be seen this stress plays the role of a 'back', 'shift' or 'drag' stress in simple kinematic hardening models. It can also arise in certain isotropic models, as will be seen in § 6 in the discussion of critical state theories.

By performing the Legendre transformation the corresponding form of the Helmholtz free energy for a decoupled material is found to be

$$F(\varepsilon_{ij}, \alpha_{ij}) = F_1(\varepsilon_{ij} - \alpha_{ij}) - G_2(\alpha_{ij}) \tag{3.9}$$

where F_1 is the Legendre dual of $G_1(\sigma_{ij})$. In a sense the function G is only a partial stress potential, since although it depends on σ_{ij} it also depends on the plastic strain α_{ij}. The 'complete' Gibbs function can be found by effecting a further Legendre transformation on G, interchanging the roles of α_{ij} and χ_{ij}, as explained in the appendix, giving

$$G^*(\sigma_{ij}, \chi_{ij}) = -G_1(\sigma_{ij}) + F_2(\chi_{ij} - \sigma_{ij}) \tag{3.10}$$

where F_2 is the Legendre dual of $G_2(\alpha_{ij})$.

If the material is linearly elastic with a stress-free reference state,

$$G_1(\sigma_{ij}) = \tfrac{1}{2} C_{ijkl}\sigma_{ij}\sigma_{kl} \tag{3.11}$$

and

$$F_1(\varepsilon^{\mathrm{e}}_{ij}) = \tfrac{1}{2} D_{ijkl}\varepsilon^{\mathrm{e}}_{ij}\varepsilon^{\mathrm{e}}_{kl} \tag{3.12}$$

where C_{ijkl} and D_{ijkl} are the constant compliance and modulus matrices, respectively. If in addition the material is isotropic then

$$G_1(\sigma) = \frac{1}{18K}(\mathrm{tr}(\sigma))^2 + \frac{1}{4G}(\mathrm{tr}(s^2)) \tag{3.13}$$

and

$$F_1(\varepsilon^{\mathrm{e}}) = \tfrac{1}{2} K(\mathrm{tr}(\varepsilon^{\mathrm{e}}))^2 + G(\mathrm{tr}(\gamma^{\mathrm{e}2})) \tag{3.14}$$

where K and G are the bulk and shear moduli, and s and γ are the deviatoric-stress and distortional-strain tensors, respectively. (If the reference state is non-zero, but at a stress σ^0_{ij}, the total stress σ_{ij} must be replaced by $\sigma_{ij} - \sigma^0_{ij}$ in the above.)

(*b*) *Coupled materials*

Although irreversible plastic deformation is of more importance for geomaterials than elastic-plastic coupling (dependence of the elastic moduli on plastic deformation), we shall pursue here coupling phenomena first, since these depend on the form of the Gibbs free-energy function, and can be addressed without yet introducing the dissipation function.

The most common example of a coupled material is one in which the elastic moduli vary with the internal parameter, in which case the above proof of the decomposition of the strain into an elastic and plastic part is no longer valid. Such models are frequently used to describe the elastic plastic behaviour of rocks and concrete, whilst many critical state soil models involve stiffnesses which depend on the specific volume and hence on the plastic volume strain (Wood 1990). Maier & Hueckel (1977) have given a comprehensive discussion of the elastic/plastic behaviour of such materials including the effects of softening and non-associated flow rules, but in the context of the classical formulation of elastic/plastic constitutive laws. Houlsby (1982) has demonstrated the way in which coupling can alter the thermomechanical formulation by considering the particular case of an isotopic material in which the shear modulus depends on the plastic strain.

The total strain rate is still given by (3.1) but we cannot now identify the two terms on the left-hand side as 'elastic' and 'plastic' strain rates. Instead Maier & Hueckel

(1977) refer to them as the 'reversible' and 'irreversible' components, respectively,

$$\dot{\varepsilon}^{\mathrm{r}}_{ij} = \frac{\partial^2 G}{\partial\sigma_{ij}\partial\sigma_{kl}}\dot{\sigma}_{kl} \quad \text{and} \quad \dot{\varepsilon}^{\mathrm{i}}_{ij} = \frac{\partial^2 G}{\partial\sigma_{ij}\partial\alpha_{kl}}\dot{\alpha}_{kl}. \tag{3.15}$$

One possible way of developing the theory is to define the plastic strain as that part of the strain which remains when the stress is returned to its original value, i.e.

$$\varepsilon^{\mathrm{p}}_{ij} = \left.\frac{\partial G(\sigma_{ij}, \alpha_{ij})}{\partial\sigma_{ij}}\right|_{\sigma^0_{ij}} = G^{\mathrm{p}}_{ij}(\alpha_{ij}) \tag{3.16}$$

say. This definition is consistent with that used for decoupled materials and shows that the plastic strain is independent of the current stress; thus, as above, we can still choose the internal variable α_{ij} to be the plastic strain. The elastic strain rate is hence

$$\dot{\varepsilon}^{\mathrm{e}}_{ij} = \dot{\varepsilon}_{ij} - \dot{\varepsilon}^{\mathrm{p}}_{ij} = \frac{\partial^2 G}{\partial\sigma_{ij}\partial\sigma_{kl}}\dot{\sigma}_{kl} + \frac{\partial^2 G}{\partial\sigma_{ij}\partial\alpha_{kl}}\dot{\alpha}_{kl} - \dot{\alpha}_{ij} \tag{3.17}$$

or

$$\dot{\varepsilon}^{\mathrm{e}}_{ij} = \dot{\varepsilon}^{\mathrm{r}}_{ij} + \dot{\varepsilon}^{\mathrm{c}}_{ij}, \tag{3.18}$$

where

$$\dot{\varepsilon}^{\mathrm{c}}_{ij} = \left(\frac{\partial^2 G}{\partial\sigma_{ij}\partial\alpha_{kl}} - \delta_{ik}\delta_{jl}\right)\dot{\alpha}_{kl} \tag{3.19}$$

is the 'coupled' strain rate defined by Maier & Hueckel (1977). The elastic strain rate is hence the sum of the reversible and coupled strain rates, whilst the irreversible strain rate is the sum of the plastic and coupled components.

If it is assumed that the Gibbs function is still given by (3.5), but with the 'elastic part' now dependent on α_{ij} as well as σ_{ij}, as is the case if any of the elastic moduli in (3.11)–(3.14) depend on α_{ij}, then

$$\varepsilon_{ij} = \frac{\partial G_1(\sigma_{ij}, \alpha_{ij})}{\partial\sigma_{ij}} + \alpha_{ij}. \tag{3.20}$$

It follows from $G = G_1(\sigma_{ij}) + \sigma_{ij}\alpha_{kl} + G_2(\alpha_{ij})$ that the coupled strain rate as given by equation (3.19) is

$$\dot{\varepsilon}^{\mathrm{c}}_{ij} = \frac{\partial^2 G_1}{\partial\sigma_{ij}\partial\alpha_{kl}}\dot{\alpha}_{kl}. \tag{3.21}$$

Note that the coupled strain rate is derived above from the formulation, and is not introduced as an additional assumption. Equation (3.21) reduces for a material which is linear elastic with respect to the stresses to

$$\dot{\varepsilon}^{\mathrm{c}}_{ij} = \frac{\partial C_{ijkl}}{\partial\alpha_{pq}}\sigma_{kl}\dot{\alpha}_{pq}. \tag{3.22}$$

The different ways in which the total strain may be split into either elastic and plastic components, or recoverable and irrecoverable components are illustrated in table 1.

4. The dissipation and yield functions

The formulation of the constitutive equation is now completed by specifying the dissipation function. To begin with we will consider dissipation functions which depend only on α_{ij} and $\dot{\alpha}_{ij}$. Other possible functional dependencies will be discussed

Table 1. *Alternative decomposition of strains for coupled materials*

total strain	ε_{ij}		
elastic + plastic	ε_{ij}^{e}		ε_{ij}^{p}
recoverabe + coupled + plastic	ε_{ij}^{r}	ε_{ij}^{c}	ε_{ij}^{p}
recoverabe + irrecoverable	ε_{ij}^{r}	ε_{ij}^{i}	

later. From the orthogonality postulate (2.15) it follows that the generalized stresses are given by

$$\chi_{ij} = \frac{\partial D}{\partial \dot{\alpha}_{ij}}(\alpha_{ij}\dot{\alpha}_{ij}) \tag{4.1}$$

so that, in general, a Legendre transformation of $D(\alpha_{ij}\dot{\alpha}_{ij})$ introduces a new function, $\Omega(\alpha_{ij}, \chi_{ij})$ say, with the properties that

$$\dot{\alpha}_{ij} = \frac{\partial \Omega}{\partial \chi_{ij}}(\alpha_{ij}, \chi_{ij}) \tag{4.2}$$

and

$$D(\alpha_{ij}, \dot{\alpha}_{ij}) + \Omega(\alpha_{ij}, \chi_{ij}) = \chi_{ij}\dot{\alpha}_{ij}. \tag{4.3}$$

Equation (4.2) is an evolutionary equation for the internal variables, and is the starting point of many thermomechanical theories of viscoplasticity (e.g. Rice 1971; Lubliner 1972).

In the case of rate-independent materials, however, D must be homogeneous of degree one in $\dot{\alpha}_{ij}$, since there is no characteristic time. For such functions the Legendre transformation is singular, as discussed in the appendix, where it is shown that the value of the Legendre dual of such a function is identically zero:

$$f(\alpha_{ij}, \chi_{ij}) = 0, \quad \text{say.} \tag{4.4}$$

Moreover since the transformation is not one-to-one, the dual relation to (4.1) is not unique, so that instead of having a unique expression for the time derivatives of the internal parameters as in (4.2), they are given by

$$\dot{\alpha}_{ij} = \dot{\lambda}\frac{\partial f}{\partial \chi_{ij}} \quad \text{and} \quad \frac{\partial D}{\partial \alpha_{ij}} = -\dot{\lambda}\frac{\partial f}{\partial \alpha_{ij}} \tag{4.5}$$

for some multiplier $\dot{\lambda}$. Equations (4.4) and the first of (4.5) are of course the yield condition and associated or normal flow rule, but both are expressed in generalized stress space not true stress space. The existence of the yield function is seen to be a necessary consequence of having a rate-independent dissipation function. Although this striking result has been known for some time in the French literature (see, for example, Moreau (1970), Halphen & Nguyen (1975) and other references cited in the book by Maugin (1992)), it would seem to be only relatively recently that it has become well known in the English language literature, largely due to the work of Martin, Reddy and co-workers, although it had been derived independently by Houlsby (1981, 1982). Even earlier, but more restricted, proofs have been given by Thomas (1954) and Sawczuk & Stutz (1968).

The question immediately arises 'does normality also hold in true stress space?'. To answer this question we must replace χ_{ij} in (4.4) by σ_{ij}. This is readily achieved for a decoupled material using (3.8) and a yield function in true stress space can be defined by

$$\bar{f}(\alpha_{ij}, \sigma_{ij}) = f(\alpha_{ij}, \sigma_{ij} - \rho_{ij}(\alpha_{ij})) = 0. \tag{4.6}$$

The first equation (4.5) can hence be rewritten in the standard form of a normal flow rule:

$$\dot{\alpha}_{ij} = \dot{\lambda}\frac{\partial \bar{f}}{\partial \sigma_{ij}}. \tag{4.7}$$

From (4.6) the stress variable ρ_{ij}, which derives from the dependence of the thermodynamic potentials on plastic strain, is seen to play the role of the 'back', 'shift' or 'drag' stress in kinematic hardening models of the Melan–Prager type.

For more general materials which exhibit elastic/plastic coupling we can replace the generalized stress in (4.4) using the Gibbs function and the second equation (2.13) and define the yield surface in true stress space by

$$\bar{f}(\alpha_{ij}, \sigma_{ij}) = f\left(\alpha_{ij}, \frac{\partial G}{\partial \alpha_{ij}}(\sigma_{ij}, \alpha_{ij})\right) \tag{4.8}$$

with normal derivative

$$\frac{\partial \bar{f}}{\partial \sigma_{ij}} = \frac{\partial f}{\partial \chi_{kl}}\frac{\partial^2 G}{\partial \alpha_{kl}\partial \sigma_{ij}} \tag{4.9}$$

so that the flow rule now takes the form

$$\dot{\lambda}\frac{\partial \bar{f}}{\partial \sigma_{ij}} = \frac{\partial^2 G}{\partial \alpha_{kl}\partial \sigma_{ij}}\dot{\alpha}_{kl} = \dot{\varepsilon}^{\mathrm{i}}_{ij} \tag{4.10}$$

using (4.5) and (3.15). Hence, for general coupled materials it is the irreversible part of the strain rate, rather than the plastic strain rate, which is given by the normal flow rule in true stress space. This explains why it is necessary to introduce non-associated flow rules to model such materials in the conventional formulation as developed by Maier & Hueckel (1977). The non-associated flow rule arises naturally in the present formulation however, and is fully described by the two fundamental constitutive functions—the Gibbs function and the dissipation function—there is no need to introduce a new function, such as a plastic potential.

(*a*) *Example*

Examples of the generation of constitutive laws for metals based on this general thermomechanical formulation have been given by Ziegler (1981, 1983), Houlsby (1982), Ziegler & Wehrli (1987), Martin & Reddy (1993) and Reddy & Martin (1994). Here we consider a simple example which illustrates a number of important points, particularly the manner in which isotropic and kinematic hardening enter the models, and the method by which internal kinematic constraints can be included in the general formulation.

The classical isotropic hardening Mises material can be generated by taking the Gibbs function in the form (3.13), giving linear elastic behaviour. There is no plastic term in this Gibbs function so that the back stress is zero and $\chi_{ij} = \sigma_{ij}$. The dissipation function is taken as

$$D = k(\gamma^{\mathrm{p}})\sqrt{2\dot{\gamma}^{\mathrm{p}}_{ij}\dot{\gamma}^{\mathrm{p}}_{ij}} \tag{4.11}$$

where

$$\dot{\gamma}^{\mathrm{p}}_{ij} = \dot{\alpha}_{ij} - \tfrac{1}{3}\dot{\alpha}_{kk}\delta_{ij} \tag{4.12}$$

is the distortional part of the plastic strain-rate tensor, and γ^{p} is the second invariant of the distortional plastic strain tensor γ^{p}_{ij} and k is the strength in simple shear. The plastic incompressibility condition can be introduced as a side constraint:

$$\dot{\nu}^{\mathrm{p}} = \dot{\alpha}_{kk} = 0, \tag{4.13}$$

where $\dot{\nu}^{\mathrm{p}}$ is the volumetric plastic strain rate. This constraint can be included by employing the standard procedure (e.g. Lippmann 1972) of introducing a Lagrange multiplier Λ and considering the augmented dissipation function

$$D^* = k(\gamma^{\mathrm{p}})\sqrt{2\dot{\gamma}^{\mathrm{p}}_{ij}\dot{\gamma}^{\mathrm{p}}_{ij}} + \Lambda\dot{\nu}^{\mathrm{p}}. \tag{4.14}$$

Differentiation of D^* with respect to the plastic strain rates gives the true stress components (since $\chi_{ij} = \sigma_{ij}$):

$$p = \frac{\partial D^*}{\partial \dot{\nu}^{\mathrm{p}}} = \Lambda \quad \text{and} \quad s_{ij} = \frac{\partial D^*}{\partial \dot{\gamma}^{\mathrm{p}}_{ij}} = \sqrt{2}k(\gamma^{\mathrm{p}})\frac{\dot{\gamma}^{\mathrm{p}}_{ij}}{\sqrt{\dot{\gamma}^{\mathrm{p}}_{kl}\dot{\gamma}^{\mathrm{p}}_{kl}}}. \tag{4.15}$$

As expected the mean pressure is a passive variable, undetermined by the constitutive equation. Squaring the second equation (4.15) enables the strain rate to be eliminated, generating the Mises yield condition

$$s_{ij}s_{ij} = 2k^2(\gamma^{\mathrm{p}}) \tag{4.16}$$

for an isotropically hardening material.

Similarly the Tresca criterion can be generated by applying the standard procedure to the augmented dissipation function.

$$D^* = k(\gamma^{\mathrm{p}})(|\dot{\gamma}^{\mathrm{p}}_1| + |\dot{\gamma}^{\mathrm{p}}_2| + |\dot{\gamma}^{\mathrm{p}}_3|) + \Lambda\dot{\nu}^{\mathrm{p}} \tag{4.17}$$

where $\dot{\gamma}^{\mathrm{p}}_i$ are the principal components of the plastic distortion-rate tensor. The principal components of the deviatoric stress tensor are

$$s_i = \frac{\partial D}{\partial \dot{\gamma}^{\mathrm{p}}_i} = k(\gamma^{\mathrm{p}})\,\mathrm{sgn}(\dot{\gamma}^{\mathrm{p}}_i) \tag{4.18}$$

so that

$$\sigma_1 - \sigma_2 = s_1 - s_2 = k(\mathrm{sgn}(\dot{\gamma}^{\mathrm{p}}_1) - \mathrm{sgn}(\dot{\gamma}^{\mathrm{p}}_2)) \tag{4.19}$$

which is equal to $2k$ if $\dot{\gamma}^{\mathrm{p}}_1$ is positive and $\dot{\gamma}^{\mathrm{p}}_2$ is negative and $-2k$ if $\dot{\gamma}^{\mathrm{p}}_1$ is negative and $\dot{\gamma}^{\mathrm{p}}_2$ is positive. If either of these two principal distortion rates is zero then $\sigma_1 - \sigma_2$ is undetermined. Finally, $\sigma_1 - \sigma_2 = 0$ if $\dot{\gamma}^{\mathrm{p}}_1$ and $\dot{\gamma}^{\mathrm{p}}_2$ have the same sign. This is only possible if $\dot{\gamma}^{\mathrm{p}}_3$ is non-zero and of the opposite sign to $\dot{\gamma}^{\mathrm{p}}_1$ and $\dot{\gamma}^{\mathrm{p}}_2$. It can be confirmed that these values correspond to the various surfaces which comprise the Tresca yield. The construction of the yield function is particularly straightforward in these two classical cases, but this procedure can become algebraically very complex for more complicated dissipation functions.

Certain types of kinematic hardening can be introduced into the model, by including the $G_2(\alpha_{ij})$ term in the Gibbs function as in (3.5), which introduces the non-zero back stress $\rho_{ij} = \sigma_{ij} - \chi_{ij}$. The yield condition in the Mises model now becomes

$$(s_{ij} - \rho'_{ij})(s_{ij} - \rho'_{ij}) - 2k^2(\gamma^{\mathrm{p}}) = 0, \tag{4.20}$$

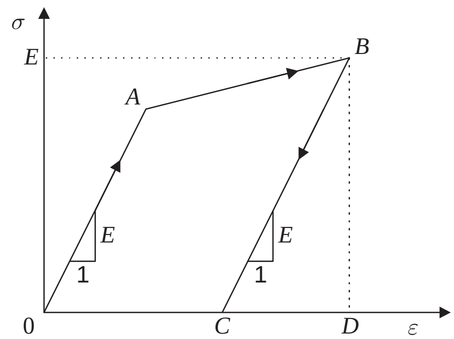

Figure 1. One dimensional elastic/plastic model using isotropic hardening.

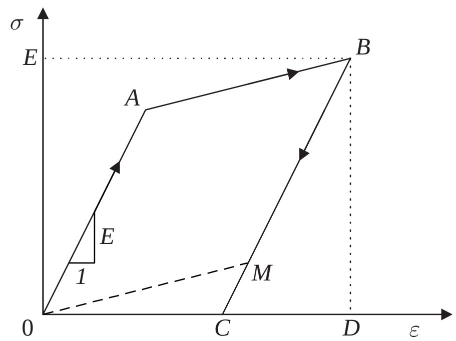

Figure 2. One-dimensional elastic/plastic model using kinematic hardening.

where ρ'_{ij} is the deviatoric part of the back-stress tensor. This is the type of kinematic hardening introduced by Melan and Prager (Lubliner 1990). Thus whilst isotropic hardening behaviour is introduced in the dissipation function, kinematic hardening arises through the thermodynamic potential function. This distinction is reflected in the decomposition of the power of deformation which can be decomposed into three terms:

$$\sigma_{ij}\dot{\varepsilon}_{ij} = \sigma_{ij}\dot{\varepsilon}^{\mathrm{e}}_{ij} + \rho_{ij}\dot{\alpha}_{ij} + \chi_{ij}\dot{\alpha}_{ij}. \tag{4.21}$$

The first two terms combine to form $\dot{F}$, the rate of change of the free energy, and the last two terms give the 'plastic work rate' but only the energy associated with the last term is dissipated. The work associated with the 'back' stress, ρ_{ij}, is hence recoverable and will be zero in a closed cycle. In contrast the work associated with the isotropic hardening component is entirely dissipated. Lubliner (1990) and Maugin (1992) discuss this issue further together with more complex forms of kinematic hardening models.

This difference between the two types of hardening can be simply illustrated by comparing the uniaxial response of a linear isotropic hardening material, with a yield stress of $\sigma = Y + h\alpha$ say, where Y is the initial yield stress and h the hardening modulus, with that of a linear kinematic hardening material, with a constant yield stress $\chi = Y$, but with a back stress $\rho = h\alpha$, so once again $\sigma = Y + h\alpha$. In both cases the tangent modulus is $hE/(h+E)$, where E is the constant Young's modulus. These responses are illustrated in figures 1 and 2, and the important variables summarized in table 2. In the isotropic hardening example the total energy dissipated is $\int \chi\,\mathrm{d}\alpha$ and is equal to the area OABCO, but in the corresponding diagram for kinematic

Table 2. *Forms of functions for one-dimensional plasticity models*

	isotropic hardening	kinematic hardening
Helmholtz free energy F	$\frac{1}{2}E(\varepsilon-\alpha)^2$	$\frac{1}{2}E(\varepsilon-\alpha)^2+\frac{1}{2}h\alpha^2$
Gibbs free energy G	$\frac{1}{2}(\sigma^2/E)+\sigma\alpha$	$\frac{1}{2}(\sigma^2/E)+\sigma\alpha-\frac{1}{2}h\alpha^2$
dissipation function D	$(Y+h\alpha)\lvert\dot{\alpha}\rvert$	$Y\lvert\dot{\alpha}\rvert$
generalized stress χ	$(Y+h\alpha)\,\mathrm{sgn}(\dot{\alpha})$	$Y\,\mathrm{sgn}(\dot{\alpha})$
true stress σ	$E(\varepsilon-\alpha)$	$E(\varepsilon-\alpha)$
back stress ρ	0 (so that $\sigma=\chi$)	$h\alpha$ (so that $\sigma=\chi+h\alpha$)
tangent modulus	$hE/(h+E)$	$hE/(h+E)$
areas in figures 1 and 2: F	BDC	OMBDO
areas in figures 1 and 2: G	OEBCO	OEBMO
areas in figures 1 and 2: D	OABCO	OABMO

hardening it is just OABMO. The difference OMC $=\int\rho\,\mathrm{d}\alpha=\frac{1}{2}h\alpha^2$ is now part of the free energy.

5. Stress-dependent dissipation functions of the form $D(\sigma_{ij},\alpha_{ij},\dot{\alpha}_{ij})$

In sections concerned with modelling the behaviour of soils and sands it will be assumed that the pore pressures are subtracted out and the stresses in the constitutive equations are the effective stresses. Houlsby (1979) showed that under the usual assumptions in soil mechanics it is the effective stress which is work conjugate to the soil skeleton strain.

The dissipation function is normally regarded as being a function of kinematic variables and so for elastic/plastic materials its most general form would be $D(\varepsilon_{ij},\alpha_{ij},\dot{\alpha}_{ij})$. Since ε_{ij} can, in general, be expressed in terms of σ_{ij} and α_{ij} through the Gibbs free-energy function, we could equally well consider the dissipation function to be of the form $D(\sigma_{ij},\alpha_{ij},\dot{\alpha}_{ij})$. This form is of particular importance in soil mechanics. Models representing the plastic behaviour of cohesionless materials have no defining parameters with the dimensions of stress, and yet the dissipation function must have dimensions of (stress)(time)$^{-1}$. Thus either the dissipation function contains one or more of the stress components explicitly, which is the case for frictional materials, or it contains a function, which has the dimensions of stress, whose physical interpretation can only be interpreted in terms of the particular model involved. This latter situation arises in the critical state models, the relevant function being related to the normal preconsolidation pressure. Examples of this latter class of theories are discussed in the next section. Here we first investigate some general properties of models with stress-dependent dissipation functions.

Applying the orthogonality principle gives the standard expression for the generalized stress:

$$\chi_{ij}=\frac{\partial D}{\partial\dot{\alpha}_{ij}}(\sigma_{ij},\alpha_{ij},\dot{\alpha}_{ij}) \tag{5.1}$$

but where the σ_{ij} terms in the dissipation function are treated as independent constant parameters when performing the differentiations. The degenerate Legendre transformation, provides the dual yield function

$$f(\sigma_{ij}, \alpha_{ij}, \chi_{ij}) = 0, \tag{5.2}$$

with generalized normal flow rule

$$\dot{\alpha}_{ij} = \dot{\lambda}\frac{\partial f}{\partial \chi_{ij}} \tag{5.3}$$

and in addition

$$\frac{\partial D}{\partial \sigma_{ij}} = -\dot{\lambda}\frac{\partial f}{\partial \sigma_{ij}} \tag{5.4 a}$$

$$\frac{\partial D}{\partial \alpha_{ij}} = -\dot{\lambda}\frac{\partial f}{\partial \alpha_{ij}} \tag{5.4 b}$$

since σ_{ij} and α_{ij} are both passive variables in this transformation (cf. the discussion of the singular transformation in the appendix; in particular equations (A 18)).

As in the previous section we can now transform the yield function (5.2) from generalized stress space to true stress space and define a yield function in this space by

$$\bar{f}(\sigma_{ij}, \alpha_{ij}) = f\left(\sigma_{ij}, \alpha_{ij}, \frac{\partial G}{\partial \alpha_{ij}}(\sigma_{ij}, \alpha_{ij})\right) \tag{5.5}$$

which has the associated normal derivative

$$\frac{\partial \bar{f}}{\partial \sigma_{ij}} = \frac{\partial f}{\partial \sigma_{ij}} + \frac{\partial f}{\partial \chi_{kl}}\frac{\partial^2 G}{\partial \alpha_{ij}\partial \sigma_{kl}}. \tag{5.6}$$

Upon multiplying by $\dot{\lambda}$ and using (3.15), the flow rule in true stress space becomes

$$\dot{\varepsilon}^{\mathrm{i}}_{ij} = \dot{\lambda}\frac{\partial \bar{f}}{\partial \sigma_{ij}} + \frac{\partial D}{\partial \sigma_{ij}} \tag{5.7}$$

which generalizes (4.9) to the case of stress-dependent dissipation functions. The irreversible strain rate reduces to the plastic strain rate when the material is decoupled. This result demonstrates that, in general, *when D depends explicitly on the stress*, the normality rule *cannot be expected to hold* in true stress space. As for the case of coupled materials, it is to be emphasized that this non-associated behaviour arises naturally from the general thermomechanical framework and no new function needs to be introduced to describe the constitutive equations.

As an example, consider the Drucker–Prager frictional model for cohesionless soils (Drucker & Prager 1952). This can be derived from a simple dissipation function:

$$D = \frac{\mu}{\sqrt{3}}p\sqrt{2\dot{\gamma}^{\mathrm{p}}_{ij}\dot{\gamma}^{\mathrm{p}}_{ij}} \tag{5.8}$$

where μ is a constant related to the friction angle and the factor $\sqrt{3}$ is introduced for consistency with notation commonly used in soil mechanics. As a side condition we will impose a linear relationship between the volumetric and shear strain rates

$$\dot{\nu}^{\mathrm{p}} + \frac{\beta}{\sqrt{3}}\sqrt{2\dot{\gamma}^{\mathrm{p}}_{ij}\dot{\gamma}^{\mathrm{p}}_{ij}} = 0, \tag{5.9}$$

where β is a constant, related to the dilation angle (this extends the Drucker–Prager

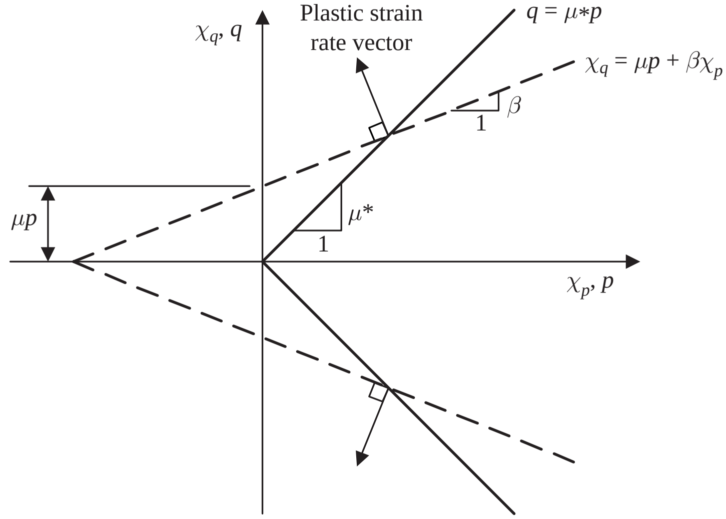

Figure 3. Yield surface and plastic potential for frictional material.

model, in which the angle of dilation was required to be the same as the angle of friction). The introduction of the constraint is a device previously used to introduce dilation into the model (Chandler 1985; Houlsby 1981, 1993). The standard procedure can now be applied to the modified dissipation function

$$D^* = \frac{\mu p + \Lambda\beta}{\sqrt{3}} \sqrt{2\dot{\gamma}_{ij}^{\mathrm{p}}\dot{\gamma}_{ij}^{\mathrm{p}}} + \Lambda\dot{\nu}^{\mathrm{p}} \tag{5.10}$$

with Λ as the Lagrange multiplier. Differentiating with respect to the plastic strain rates, gives the components of the generalized stress tensor

$$\chi_p = \Lambda \quad \text{and} \quad \chi'_{ij} = \sqrt{\tfrac{2}{3}}(\mu p + \Lambda\beta)\frac{\dot{\gamma}_{ij}^{\mathrm{p}}}{\sqrt{\dot{\gamma}_{ij}^{\mathrm{p}}\dot{\gamma}_{ij}^{\mathrm{p}}}} \tag{5.11}$$

where $\chi_p = \frac{1}{3}\chi_{ij}$, and χ'_{ij} is the deviatoric part of χ_{ij}. Eliminating the plastic strain rates between equations (5.11), generates the yield surface in (χ_p, χ_q) space:

$$|\chi_q| = \mu p + \beta\chi_p \tag{5.12}$$

where

$$\chi_q = \sqrt{\tfrac{3}{2}\chi'_{ij}\chi'_{ij}}. \tag{5.13}$$

This yield function in generalized stress space is of Coulomb type with an apparent cohesion of μp, represented by two straight lines of slope as shown in figure 3. The strain-rate vector is orthogonal to this yield surface from (5.9) and (5.12) so that the normality rule holds in this space as predicted by (5.3).

If now we transform this yield condition to true stress space, putting $\chi_p = p$, $\chi_q = q$, which presumes that the Gibbs potential is such that there is no back stress, then (5.12) becomes

$$q = \mu^* p, \quad \text{where } \mu^* = \mu + \beta \tag{5.14}$$

which is the Drucker–Prager generalization of the classical Coulomb yield condition for a cohesionless soil. We now only have normality if $\mu = 0$ and $\mu^* = \beta$ in which case the dissipation is zero. The material is incompressible when $\beta = 0$. The effect of including back stresses is to translate the yield lines in q–p space and has no effect on normality.

In order to understand further the extra complexities introduced by allowing the dissipation function to depend on stress, it is useful to consider one of the geometric interpretations of the Legendre transformation. These are described more fully in the appendix. The function ϕ is defined by

$$\phi(\sigma_{ij}, \alpha_{ij}, \dot{\alpha}_{ij}) \equiv \sigma_{ij}\dot{\alpha}_{ij} - D(\sigma_{ij}, \alpha_{ij}, \dot{\alpha}_{ij}) = 0. \tag{5.15}$$

For simplicity the material is assumed to be decoupled with zero backstress, so that ϕ is identically zero. Keeping the plastic strain constant, (5.15) defines a surface, S say, in true stress space for each direction of the plastic strain rate $\dot{\alpha}_{ij}$. As $\dot{\alpha}_{ij}$ varies over all possible values the family of surfaces S envelope a surface, Σ say, which encloses that region in stress space which cannot be attained by any plastic deformation, i.e. the yield surface. This geometrical view point of relating the dissipation function and the yield surface was used by Chandler (1985) in his discussion of stress-dependent dissipation functions. Chandler's arguments are similar to those presented here except that he did not consider a complete thermodynamic model, nor make explicit use of Legendre transformations. The equation of the yield surface envelope is obtained by eliminating the $\dot{\alpha}_{ij}$ values between (5.15) and

$$\frac{\partial \phi}{\partial \dot{\alpha}_{ij}} = \sigma_{ij} - \frac{\partial D}{\partial \dot{\alpha}_{ij}} = 0, \tag{5.16}$$

which is obtained by differentiating (5.15) with respect to the plastic strain rates. Equation (5.16) is just a restatement of the orthogonality principle (2.15) when the back stress is zero. Furthermore the normals to the yield surface Σ and to a surface S must be in the same direction at their common point of tangency, so

$$\dot{\lambda}\frac{\partial \bar{f}}{\partial \sigma_{ij}} = \frac{\partial \phi}{\partial \sigma_{ij}} = \dot{\alpha}_{ij} - \frac{\partial D}{\partial \sigma_{ij}} \tag{5.17}$$

which is the flow rule (5.7) for a decoupled material.

When the dissipation function D does *not* depend explicitly on the stress the surfaces S defined by (5.15) are hyperplanes, whose normals are in the direction of $\dot{\alpha}_{ij}$ and which are a distance D/α_m from the origin, where α_m is a suitably-defined tensor norm. The normals to the enveloped yield surface Σ must also be in the direction of the $\dot{\alpha}_{ij}$s, so that the flow rule is necessarily associated.

For an isotropic material the function ϕ depends on the stress and plastic strain rate only through their invariants and joint invariants, which can most conveniently be written $\mathrm{tr}(\sigma^m)$, $\mathrm{tr}(\dot{\alpha}^m)$ for $m = 1, 2, 3$ and $\mathrm{tr}(\sigma^p \dot{\alpha}^q)$ for $p, q = 1, 2$, (cf. Baker & Desai 1984; Prevost 1987). In addition, this function must be first order in $\dot{\alpha}_{ij}$: it must also be first order in σ_{ij}, in the case of a 'purely frictional' material which does not have any defining material parameters with the dimensions of stress. The Drucker–Prager model provides one example. An alternative is given by the dissipation function,

$$D = \mu\sqrt{\tfrac{8}{9}(\mathrm{tr}(\sigma)\,\mathrm{tr}(\sigma\dot{\alpha}^2) - (\mathrm{tr}(\sigma\dot{\alpha}))^2)} \tag{5.18}$$

which generates the Matsuoka–Nakai (1974) yield condition:

$$\frac{I_1 I_2}{I_3} \equiv \mathrm{tr}(\sigma)\,\mathrm{tr}(\sigma^{-1}) = \mu^2 \tag{5.19}$$

where $I_i = 1, 2, 3$ are the three standard stress invariants. For such a model the equations $\phi = 0$ generate a family of *non-planar* surfaces in stress space. This is in marked contrast to stress-independent dissipation functions of the classical theory, which always produce families of hyperplanes, as noted above. Since a frictional material does not have any material parameter with the dimensions of stress, the yield function can always be written as a homogeneous function of degree zero in the stress components and the corresponding yield surface must be a generalized cone with vertex at the origin.

A given yield surface Σ can be generated from an infinity of families of enveloping surfaces S, corresponding to the infinity of flow rules that can be generated from (5.17). However, a prescribed yield surface can only be generated from *one* family of tangential hyperplanes. For an isotropic material the equations of this family of hyperplanes must take the form

$$\phi(\sigma, \dot{\alpha}) = \mathrm{tr}(\sigma\dot{\alpha}) - \mathrm{tr}(\sigma)\,\Psi_1(\dot{\alpha}) - \sigma_0\,\Psi_2(\dot{\alpha}) \tag{5.20}$$

where Ψ_1 and Ψ_2 are first-order functions of the three invariants of $\dot{\alpha}_{ij}$, and σ_0 is zero for a frictional material and is otherwise some material-characterizing stress parameter. Differentiating (5.20) and substituting in (5.17), the difference between the plastic strain rate and the normal to the yield surface is seen to be $\Psi_1(\dot{\alpha})\delta_{ij}$, an isotropic tensor, representing a dilational strain rate whose components are independent of stress. For isotropic materials therefore, a linear dissipation function always produces a flow rule which is associated in the Π plane (i.e. the deviatoric part of the plastic strain rate is normal to the cross-section of the yield surface in a Π-plane) and the lack of normality occurs only in the dilational part of the strain rate. This result has been derived before by Sawczuk & Stutz (1968) in a more restricted context, but does not seem well known in the soil-mechanics literature. Many experimenters have shown that non-associated behaviour is predominately of this type (e.g. Lade & Kim 1988) and normality in the Π-plane is a common assumption in many mathematical models (e.g. Prevost 1987). Because of the algebraic complexity of performing the Legendre transformation in all cases other than when the functions are quadratic, the actual forms of the functions in (5.20) will normally be very complicated, even when the expression for the yield function in terms of the stress invariants is algebraically simple.

We hence conclude that for an isotropic material it is always possible to associate a linear dissipation function, which necessarily produces a flow rule which has normality in the deviatoric plane, with any given yield surface; but this does not exclude the possibility of producing nonlinear dissipation functions with more general forms of non-associated behaviour.

6. Critical state models

Elastic–plastic models using critical state formulations have been very successful in describing many of the more important aspects of the mechanical behaviour of soils, particularly for normally consolidated clays. The theories are normally developed in terms of work-conjugate variables, whose value can be determined from triaxial tests, namely the volumetric and effective shear strains, ν and γ, the mean effective pressure p, and the principal stress difference q.

The dissipation function for such a material can be written in the form

$$D = \Pi(\nu^{\mathrm{p}})T(\dot{\nu}^{\mathrm{p}}, \dot{\gamma}^{\mathrm{p}}) \tag{6.1}$$

where ν^{p} is the plastic volume strain, and $\dot{\nu}^{\mathrm{p}}, \dot{\gamma}^{\mathrm{p}}$, are the plastic components of the volumetric and shear strain rates. The function Π, which has the dimensions of stress, defines the volumetric hardening behaviour, whilst T, which has dimensions of $(\mathrm{time})^{-1}$, governs the energy dissipation rate and hence determines the shape of the yield surfaces. The form of these two functions for the original and Modified Cam–Clay have already found by Modaressi *et al.* (1994) and Houlsby (1981), respectively, and are given here in table 2. Assuming a decoupled model the Gibbs function must be of the form

$$G(p, q, \nu^{\mathrm{p}}) = G_1(p, q) + p\nu^{\mathrm{p}} + q\gamma^{\mathrm{p}} + G_2(\nu^{\mathrm{p}}, \gamma^{\mathrm{p}}) \tag{6.2}$$

from (3.5). The form (3.13) for the function G_1 governing the elastic response would give constant elastic moduli. However, for soils the instantaneous bulk modulus is not constant and is effectively proportional to the mean effective pressure, which requires G_1 to be of the form

$$G_1(p, q) = \kappa^* p(\ln p - 1) + \frac{q^2}{6G} \tag{6.3}$$

where the shear modulus is still assumed to be constant. The stress strain relations are

$$\nu^{\mathrm{e}} = \kappa^* \ln p \quad \text{and} \quad \gamma^{\mathrm{e}} = \frac{q}{3G} \tag{6.4}$$

and

$$\dot{\nu}^{\mathrm{e}} = \kappa^* \frac{\dot{p}}{p} \quad \text{and} \quad \dot{\gamma}^{\mathrm{e}} = \frac{\dot{q}}{3G}. \tag{6.5}$$

In most formulations of critical state theories, the instantaneous bulk modulus also depends on the current specific volume V or voids ratio e (Wood 1990), and the first equation (6.5) is replaced by

$$\dot{\nu}^{\mathrm{e}} = \frac{\kappa}{1+e} \frac{\dot{p}}{p}. \tag{6.6}$$

Since e depends on the plastic strain, such a model is 'coupled' and introduces all the complexities discussed in § 3. This coupling, however, is very weak and is in any case second order in a small strain theory. Furthermore, Butterfield (1979) has demonstrated that in most cases the experimental data is equally well represented by the relations (6.4) or (6.5), which predict linear relations between $\ln V$ and $\ln p$, as by (6.6) in which V and $\ln p$ are linearly related.

Since $T(\dot{\nu}^{\mathrm{p}}, \dot{\gamma}^{\mathrm{p}})$ is homogeneous of degree one in the strain rates it has the degenerate Legendre dual $f(x, y) = 0$ say, where $x = \partial T/\partial \dot{\nu}^{\mathrm{p}}$ and $y = \partial T/\partial \dot{\gamma}^{\mathrm{p}}$. It follows from (2.15) and (6.1) that $x = \chi_p/\Pi$ and $y = \chi_q/\Pi$, where χ_p and χ_q are the first two invariants of the generalized stress tensor. Hence, the yield condition is necessarily of the form

$$f\left(\frac{\chi_p}{\Pi(\nu^{\mathrm{p}})}, \frac{\chi_q}{\Pi(\nu^{\mathrm{p}})}\right) = 0. \tag{6.7}$$

The yield surfaces, at constant plastic volume strain, in generalized stress space are hence geometrically self-similar curves. In true stress space the yield function is

$$f\left(\frac{p-\rho_p}{\Pi(\nu^{\mathrm{p}})}, \frac{q-\rho_q}{\Pi(\nu^{\mathrm{p}})}\right) = 0, \tag{6.8}$$

Table 3. p_0 *is normal preconsolidation pressure*

	original Cam–Clay	Modified Cam–Clay
$\Pi(\nu^{\mathrm{p}})$	$p_0(\nu^{\mathrm{p}})$	$\frac{1}{2}p_0(\nu^{\mathrm{p}})$
$T(\dot{\nu}^{\mathrm{p}}, \dot{\gamma}^{\mathrm{p}})$	$M\dot{\gamma}^{\mathrm{p}}\exp((\dot{\nu}^{\mathrm{p}}/M\dot{\gamma}^{\mathrm{p}})-1)$	$\sqrt{\dot{\nu}^{\mathrm{p}2}+M^2\dot{\gamma}^{\mathrm{p}2}}$
ρ_p	0	$\frac{1}{2}p_0(\nu^{\mathrm{p}})$
p	$p_0T/M\dot{\gamma}^{\mathrm{p}}$	$\frac{1}{2}p_0(1+\dot{\nu}^{\mathrm{p}}/T)$
q	$p_0T(1-\dot{\nu}^{\mathrm{p}}/M\dot{\gamma}^{\mathrm{p}})/\dot{\gamma}^{\mathrm{p}}$	$\frac{1}{2}p_0M^2\dot{\gamma}^{\mathrm{p}}/T$
yield condition	$q = Mp\ln(p_0/p)$	$q^2 = M^2p(p_0-p)$
flow rule	$\dot{\nu}^{\mathrm{p}}/\dot{\gamma}^{\mathrm{p}} = M-(q/p)$	$\dot{\nu}^{\mathrm{p}}/\dot{\gamma}^{\mathrm{p}} = M^2(p^2-q^2)/2pq$

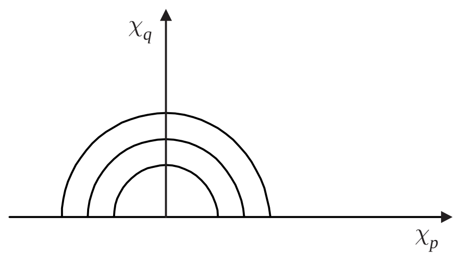

Figure 4. Yield surfaces in generalized stress space.

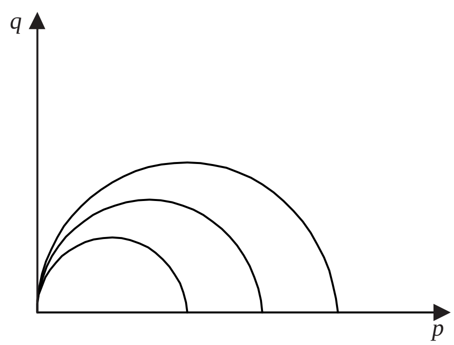

Figure 5. Yield surfaces in stress space.

where ρ_p and ρ_q are the first two invariants of the back stress tensor, given by differentiating G_2 in (6.2).

As can be seen from table 3 the original Cam–Clay model can be generated without the need to introduce a back stress and so all the plastic work is dissipated. To generate the modified Cam–Clay constitutive equation, however, Houlsby (1981) introduced a back stress of $\rho_p = \frac{1}{2}p_0(\nu^{\mathrm{p}}) = \Pi(\nu^{\mathrm{p}})$ which has the effect of shifting the self-similar yield loci in generalized stress space so that they all pass through the origin in true stress space (figures 4 and 5). However, it is interesting to note that the same yield surface and flow rule can formally be generated by adding a term $\frac{1}{2}p_0(\nu^{\mathrm{p}})\dot{\nu}^{\mathrm{p}}$ to the dissipation function, so that

$$D = \tfrac{1}{2}p_0(\nu^{\mathrm{p}})\left(\sqrt{\dot{\nu}^{\mathrm{p}2}+M^2\dot{\gamma}^{\mathrm{p}2}}+\dot{\nu}^{\mathrm{p}}\right) \tag{6.9}$$

and making G_2, and hence ρ_p zero. This modified dissipation function certainly satisfies the basic requirement that $G \geqslant 0$; however, the term added into (6.9) is integrable and hence contributes nothing to the work expanded in a closed cycle. The decomposition of the power of dissipation into the rate of change of free energy ($\dot{F}$) and dissipation (D) in (2.7′), is hence not unique, unless we also insist that D has no integrable components. In any event whichever formalism is used, it follows that in the Modified Cam–Clay model only half the work done in isotropic plastic compression is dissipated. Whilst the original Cam–Clay formulation predicts that all the plastic work is dissipated, this aspect of this model is unsatisfactory, since the associated flow rule actually predicts plastic shear strains during isotropic compression ($q = 0$), and it is necessary to impose a discontinuity on this edge of the state boundary surface. It was, of course, this feature which led to the development of the modified model.

7. Conclusions

In this paper we have developed a plasticity formulation appropriate for the development of constitutive models for granular materials. The special features of the approach are first that it is based on thermodynamic principles, so that all models developed within the framework automatically obey thermodynamic laws. Second it is sufficiently flexible to accommodate features such as friction and dilation, which are essential if geomaterials are to be modelled with reasonable accuracy.

The approach studied here begins with functions of kinematic variables, but since these are rather unfamiliar to many of those accustomed to the conventional approaches to plasticity of soils, the formulation is recast into stress variables. This reformulation proves to be very valuable, since it reveals important connections between the dissipation function, yield surface and flow rule. In particular the concepts of friction and of non-associated flow are seen to be intimately linked. The paper demonstrates that elastic/plastic models with non-associated flow rules can be incorporated in a sound theoretical framework without violating any thermodynamic laws.

The use of dissipation functions, and their connection with the form of the yield surface, is of application in some modern approaches to material behaviour in which phenomenological models are deduced from consideration of microstructural models in which dissipation may be estimated.

Examples have been provided illustrating the use of the new formulation for both uncoupled and coupled materials, and for plasticity models involving work hardening, friction and dilation. The Cam–Clay family of materials can also be accommodated within the new approach.

This paper has made much use of the Legendre transformation, which proves to be a powerful technique in the development of constitutive models from thermodynamic functions.

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Appendix: Legendre transformations

The Legendre transformation is one of the most useful in applied mathematics, although its role is not always explicitly recognized. Well-known examples include the

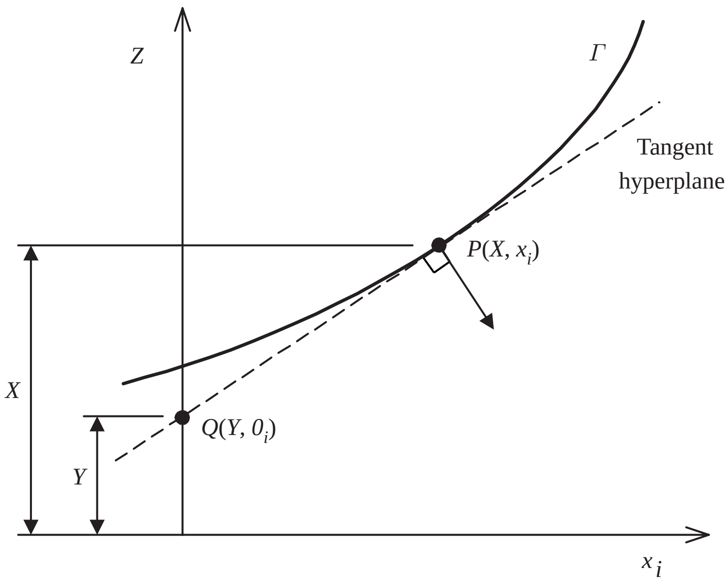

Figure 6. Representation of Γ in $(n+1)$-dimensional space.

relation between the Lagrangean and Hamiltonian functions in analytical mechanics, between strain energy and complementary energy in elasticity theory, between thermodynamic potentials and between the physical and hodograph planes in the theories of the flow of fluids and plastic solids. Sewell (1987) presents a comprehensive account of the Legendre transformation in the general theory of complementary variational and extremum principles, with particular emphasis on singular points. The transformation has also been used in rate formulations of elastic/plastic materials to transfer between stress-rate and deformation-rate potentials (see, for example, Hill 1959, 1978, 1987; Sewell 1987). These applications are rather different from those used here. We review therefore those basic properties of the transformation which are needed in the text.

(*a*) *Geometrical representation of the Legendre transformation*

A function $Z = X(x_i)$, $i = 1, 2, \ldots, n$, defines a surface, Γ, in $(n+1)$-dimensional (Z, x_i) space. The same surface can be regarded as the envelope of tangent hyperplanes. One approach to the Legendre transformation is that it allows one to construct the functional representation which describes Z in terms of these hyperplanes. This relationship is a well-known duality in geometry. The gradients of the function $X(x_i)$ are denoted by y_i:

$$y_i = \frac{\partial X}{\partial x_i} \tag{A 1}$$

so that the normal to Γ in the $(n+1)$-dimensional space is $(-1, y_i)$. If the tangent hyperplane at the point $P(X, x_i)$ on Γ cuts the Z-axis at $Q(-Y, 0_i)$, the vector $(X+Y, x_i)$ lies in the tangent hyperplane (figure 6), and hence is orthogonal to the normal to Γ at P. The scalar product of these two vectors therefore yields

$$X(x_i) + Y(y_i) = x_i y_i. \tag{A 2}$$

The function $Z = -Y(y_i)$ defines the family of enveloping tangent hyperplanes and is hence the dual description of the surface Γ. The function can be found by eliminating the n variables x_i from the $(n+1)$ equations in (A 1) and (A 2). This is possible locally, providing (A 1) can be inverted and solved for the x_is, i.e. provided the Hessian matrix, $(\partial y_i/\partial x_j) = (\partial^2 X/\partial x_i \partial x_j)$, is non-singular. Points at which the determinant of the Hessian matrix vanishes are singularities of the transformation (Sewell 1987). Differentiating (A 2) at a non-singular point with respect to y_i gives

$$\frac{\partial X}{\partial x_j}\frac{\partial x_j}{\partial y_i} + \frac{\partial Y}{\partial y_i} = y_j \frac{\partial x_j}{\partial y_i} + x_i \tag{A 3}$$

which, by virtue of (A 1) reduces to

$$x_i = \frac{\partial Y}{\partial y_i}. \tag{A 4}$$

Relations (A 1), (A 2) and (A 4) define the Legendre transformation. This transformation is self-dual since the roles of X and Y can be interchanged.

The transformation is not in general readily performed analytically. An exception is when $X(x_i)$ is a quadratic form, $X(x_i) = \frac{1}{2}A_{ij}x_i x_j$, where A_{ij} is a non-singular symmetric matrix. The dual variables are hence $y_i = (\partial X/\partial x_i) = A_{ij}x_j$, so that $x_i = A_{ij}^{-1}y_j$ and the Legendre dual is also a quadratic form,

$$Y(y_i) = x_i y_i - X(x_i) = A_{ij}^{-1}y_i y_j - \tfrac{1}{2}A_{ij}^{-1}y_i y_j = \tfrac{1}{2}A_{ij}^{-1}y_i y_j. \tag{A 5}$$

Note that some authors change the sign of the function Y; the notation above is used because it reveals the symmetry of the transformation most clearly. Sewell (1987) gives an alternative geometric interpretation of the transformation.

(b) *Homogeneous functions*

Of particular importance are cases where the function $Z = X(x_i)$ is homogeneous of degree n in the x_is, so that $X(\lambda x_i) = \lambda^n X(x_i)$ for any scalar λ. From Euler's theorem it follows that:

$$nX(x_i) = x_i \frac{\partial X}{\partial x_i} = x_i y_i, \tag{A 6}$$

so that from (A 2)

$$mY(y_i) = x_i y_i = y_i \frac{\partial Y}{\partial y_i} \tag{A 7}$$

where $(1/n) + (1/m) = 1$. So that the Legendre dual $Y(y_i)$ is homogeneous of degree $m = n/(n-1)$.

In the example above $n = 2$, so that X and Y are both homogeneous of degree two. A familiar example of this situation is in linear elasticity, where the elastic strain energy $W(\varepsilon_{ij})$ and the complementary energy $W_c(\sigma_{ij})$ are both quadratic functions of their argument and satisfy the fundamental relation:

$$W(\varepsilon_{ij}) + W_c(\sigma_{ij}) = \sigma_{ij}\varepsilon_{ij}. \tag{A 8}$$

(c) *Partial Legendre transformations and closed chains of transformations*

Suppose now the functions depend on two families of variables $X(x_i, \alpha_i)$, say where x_i and α_i are n- and m-dimensional vectors, respectively. We can perform the Legendre transformation with respect to the x_i-variables as above and obtain the dual function $Y(y_i, \alpha_i)$. The variables α_i play a passive role in this transformation and are treated as constant parameters. The three basic equations are hence now

$$X(x_i, \alpha_i) + Y(y_i, \alpha_i) = x_i y_i, \tag{A 9}$$

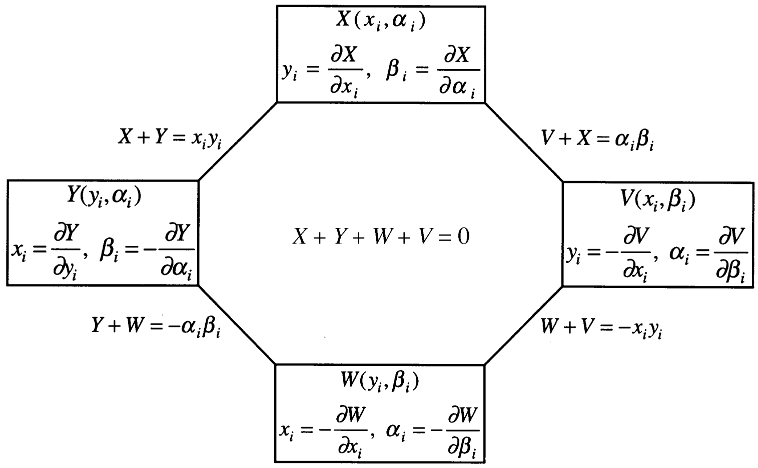

Figure 7. Chain of four Legendre transformations.

$$y_i = \frac{\partial X}{\partial x_i} \quad \text{and} \quad x_i = \frac{\partial Y}{\partial y_i}. \tag{A 10}$$

If the derivatives of X with respect to the passive variables α_i are denoted by β_i, then it follows from (A 9) that

$$\beta_i = \frac{\partial X}{\partial \alpha_i} = -\frac{\partial Y}{\partial \alpha_i}. \tag{A 11}$$

It is also possible to perform a Legendre transformation on $X(x_i, \alpha_i)$ with respect to the α_i variables and construct a second dual function $V(x_i, \beta_i)$ with the properties:

$$X(x_i, \alpha_i) + V(x_i, \beta_i) = \alpha_i \beta_i \tag{A 12}$$

where

$$\beta_i = \frac{\partial X}{\partial \alpha_i}, \quad \alpha_i = \frac{\partial V}{\partial \beta_i} \tag{A 13}$$

and furthermore

$$y_i = \frac{\partial X}{\partial x_i} = -\frac{\partial V}{\partial x_i} \tag{A 14}$$

since now the x_is are the passive variables.

This process can be continued. A Legendre transformation on $Y(y_i, \alpha_i)$ with respect to the α_i-variables produces a fourth function $W(y_i, \beta_i)$. The same function is obtained by transforming $V(x_i, \beta_i)$ with respect to the x_i-variables. A closed chain of transformation is hence produced as shown in figure 7, where the basic differential relations are summarized. The best known example of a closed chain is in classical thermodynamics, where the four functions are the internal energy $U(s, v)$, the Helmholtz free energy $F(\theta, v)$, the Gibbs free energy $G(\theta, p)$ and the free enthalpy $H(s, p)$, where θ, s, v and p are the temperature, entropy, specific volume and pressure, respectively, (see, for example, Callen 1960). Other examples are given by Sewell (1987).

(*d*) *The singular transformation*

An important case in rate-independent plasticity theory occurs when X is homogeneous and of degree one in x_i, i.e. $\lambda X(x_i, \alpha_i) = X(\lambda x_i, \alpha_i)$. It follows that

$X(x_i, a_i) = x_i y_i$, in which case the dual function $Y(y_i)$ is identically zero from (A 2). There is a simple geometric interpretation of this far-reaching result: the $(n+1)$-dimensional surface $Z = X(x_i, a_i)$ is a hypercone with its vertex at the origin. All the tangent hyperplanes hence meet the Z-axis at $Z = 0$, so that $Y(y_i, a_i) = 0$ for all y_i. Furthermore the value of $y_i = (\partial X/\partial x_i)$ is unaffected by the transformation $x_i \to \lambda x_i$ and so the mapping from $x_i \to y_i$ is $\infty \to 1$. Furthermore since the dual function $Y(y_i, \alpha_i)$ is identically zero the following are obtained,

$$\mathrm{d}Y = \frac{\partial Y}{\partial y_i}\,\mathrm{d}y_i + \frac{\partial Y}{\partial \alpha_i}\,\mathrm{d}\alpha_i = 0, \tag{A 15}$$

$$x_i\,\mathrm{d}y_i + y_i\,\mathrm{d}x_i = \mathrm{d}X + \mathrm{d}Y = \frac{\partial X}{\partial x_i}\,\mathrm{d}x_i + \frac{\partial X}{\partial \alpha_i}\,\mathrm{d}\alpha_i \tag{A 16}$$

which by virtue of (A 1) reduces to

$$x_i\,\mathrm{d}y_i - \frac{\partial X}{\partial \alpha_i}\,\mathrm{d}\alpha_i = 0, \tag{A 17}$$

Hence, by comparing (A 15) with (A 17), it follows that

$$x_i = \lambda\frac{\partial Y}{\partial y_i} \quad \text{and} \quad -\frac{\partial X}{\partial \alpha_i} = \lambda\frac{\partial Y}{\partial \alpha_i} \tag{A 18}$$

where λ is an undetermined scalar, reflecting the non-unique nature of this singular transformation.

The above development is classical in the sense that all the functions are assumed to be sufficiently smooth for all derivatives to exist. In practice the surfaces encountered in plasticity theory on occasions contain flats, edges and corners. Such surfaces, and the functions defining them, can be included in the general theory using the concepts of convex analysis. In particular the commonly defined derivative is replaced by the 'sub-differential' and the simple Legendre transformation is generalized to the 'Legendre–Fenchel' transformation. For simplicity we have used the classical notation in this paper. Treatments of the thermomechanics of elastic/plastic materials in the modern notation may be found in Maugin (1992) and Reddy & Martin (1994); the former contains useful appendices on the relevant aspects of convex analysis.

Since our main concern here is to examine the transformation as it effects the developments of constitutive laws, we have not highlighted the behaviour of any convexity properties of the various functions under the transformations. These considerations are important for questions of uniqueness, stability and the proof of extremum principles, which are beyond the scope of this paper but may be fruitful areas for future research. Some of these aspects of Legendre transformations are considered at length in the book by Sewell (1987).

Notation

c	cohesion
$D = \theta\dot{s}^{\mathrm{i}}$	dissipation function
e	voids ratio
$f(x_{ij}, \alpha_{ij})$	yield function in generalized stress space
$\bar{f}(\sigma_{ij}, \alpha_{ij})$	yield function in stress space
$F(\varepsilon_{ij}, \alpha_{ij}, \theta)$	Helmholtz free-energy function

$g(\sigma_{ij}, \alpha_{ij})$	plastic potential function
G	elastic shear modulus
$G(\sigma_{ij}, \alpha_{ij}, \theta)$	Gibbs free-energy function
$H(\sigma_{ij}, \chi_{ij}, \theta)$	complete Gibbs free-energy function
K	elastic bulk modulus
$p = \frac{1}{3}\sigma_{kh}$	mean pressure
$q = \sqrt{\frac{3}{2}s_{ij}s_{ij}} = \sigma_1 - \sigma_3$	principal stress difference in triaxial test and second deviatoric stress invariant.
q_k	heat flux vector
s	entropy
$s_{ij} = \sigma_{ij} - \frac{1}{3}\sigma_{kh}\delta_{ij}$	deviatoric stress tensor
$U(\varepsilon_{ij}, \alpha_{ij}, s)$	internal energy function
$\nu = \varepsilon_{kk}$	volumetric strain
$\dot{\nu} = \dot{\varepsilon}_{kk}$	volumetric strain-rate
V	specific volume
α_{ij}	internal variable tensor (plastic strain)
$\gamma = \sqrt{\frac{2}{3}\gamma_{ij}\gamma_{ij}}$	shear strain invariant in triaxial test.
$\gamma_{ij} = \varepsilon_{ij} - \frac{1}{3}\varepsilon_{kk}\delta_{ij}$	distortional (deviatoric) strain tensor
$\gamma_{ij} = \dot{\varepsilon}_{ij} - \frac{1}{3}\dot{\varepsilon}_{kk}\delta_{ij}$	distortional (deviatoric) strain-rate tensor
$\dot{\gamma} = \sqrt{\frac{2}{3}\dot{\gamma}_{ij}\dot{\gamma}_{ij}}$	shear strain-rate invariant
ε_{ij}	small strain tensor (compressive strains positive)
$\dot{\varepsilon}_{ij}$	rate of strain tensor
$\varepsilon^{\mathrm{e}}_{ij}$	elastic strain tensor
$\varepsilon^{\mathrm{p}}_{ij}$	plastic strain tensor
θ	temperature
κ	slope of elastic swelling line in critical state models
$\dot{\lambda}$	plastic multiplier
$\mathit{\Lambda}$	Lagrange multiplier
$\rho_{ij} = \sigma_{ij} - \chi_{ij}$	'back', 'shift' or 'drag' stress
χ_{ij}	generalized stress tensor

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